
Algorithm 5: Minimum Transit Time (MTT) Precomputation

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $country(v)$. Country associated with node v

Output:

- MTT . Map of each country to the absolute minimum time required to enter it

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1: Initialize  $MTT(c) \leftarrow \infty$  for all  $c \in$  unique countries
2: // Process scheduled edges
3: for all  $(v, u, dep, arr) \in E_s$  do
4:   if  $country(v) \neq country(u)$  then
5:     transit_time  $\leftarrow arr - dep$ 
6:     if transit_time  $< MTT(country(u))$  then
7:        $MTT(country(u)) \leftarrow transit\_time$ 
8: // Process transfer edges (e.g., cross-border footpaths or local transit)
9: for all  $(v, w, \Delta) \in E_t$  do
10:  if  $country(v) \neq country(w)$  then
11:    if  $\Delta < MTT(country(w))$  then
12:       $MTT(country(w)) \leftarrow \Delta$ 
13: return  $MTT$ 
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